

7. CRIPT – ARITHMETIC PROBLEM

Aim:

To solve the Crypt Arithmetic Problem $\text{SEND} + \text{MORE} = \text{MONEY}$ using the Backtracking Search algorithm in Artificial Intelligence.

Objective :

To assign a unique digit (0–9) to each letter such that the arithmetic equation

$$\text{SEND} + \text{MORE} = \text{MONEY}$$

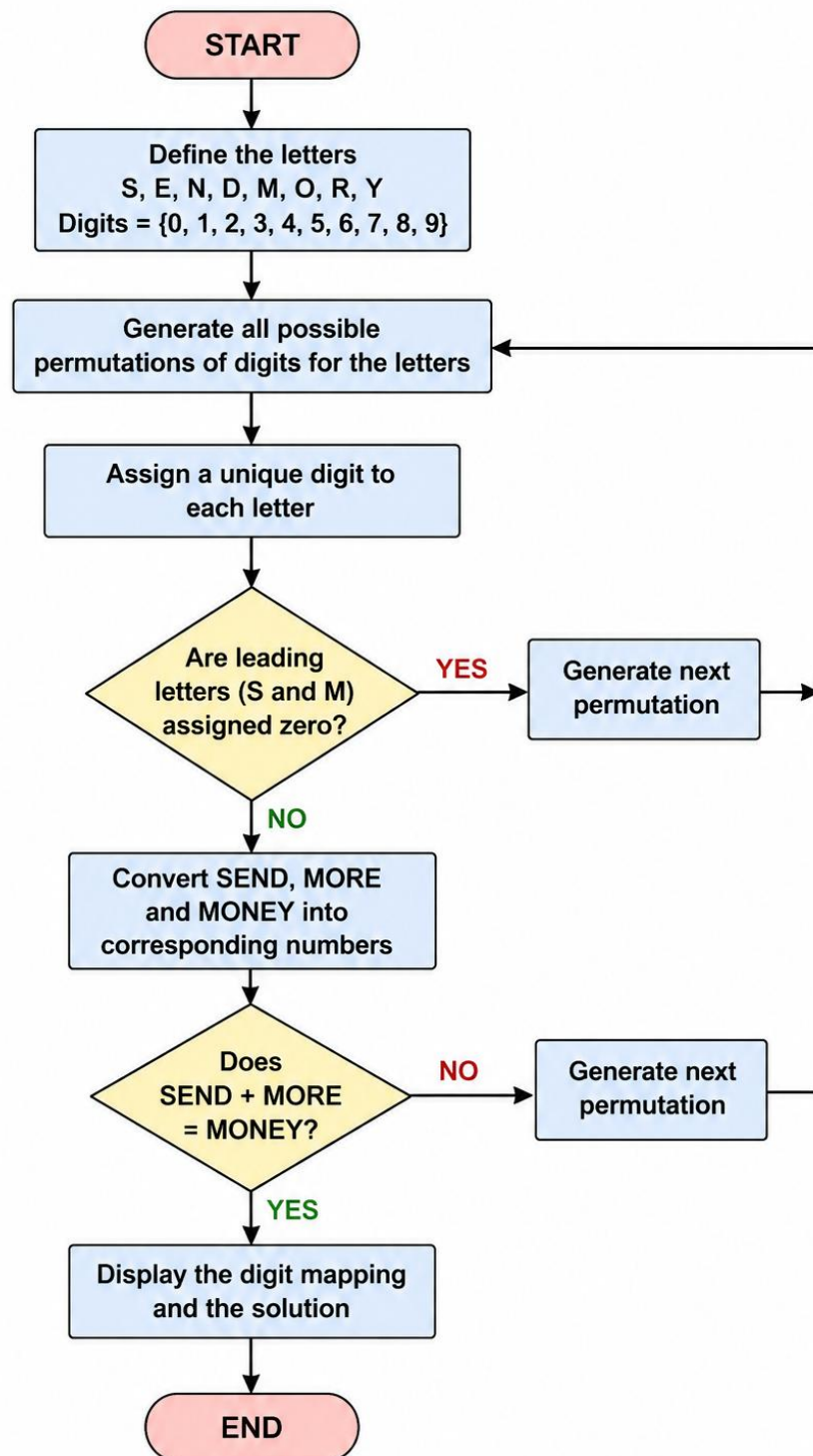
is satisfied while ensuring that:

- Each letter represents a unique digit.
- No leading letter is assigned zero.
- The equation is mathematically correct.

Algorithm:

1. Define the letters involved in the puzzle.
2. Generate all possible digit assignments using permutations.
3. Ensure that the leading letters (S and M) are not assigned zero.
4. Convert the words SEND, MORE, and MONEY into their corresponding numeric values.
5. Check whether the equation $\text{SEND} + \text{MORE} = \text{MONEY}$ is satisfied.
6. If a valid assignment is found, display the mapping and terminate.
7. Otherwise, continue searching until all possibilities are exhausted.

Flowchart :



Code:

```
from itertools import permutations

def solve_cryptarithmic():
    letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
    digits = range(10)
    for perm in permutations(digits, len(letters)):
        mapping = dict(zip(letters, perm))
        if mapping['S'] == 0 or mapping['M'] == 0:
            continue
        SEND = (mapping['S'] * 1000 +
                 mapping['E'] * 100 +
                 mapping['N'] * 10 +
                 mapping['D'])
        MORE = (mapping['M'] * 1000 +
                 mapping['O'] * 100 +
                 mapping['R'] * 10 +
                 mapping['E'])
        MONEY = (mapping['M'] * 10000 +
                  mapping['O'] * 1000 +
                  mapping['N'] * 100 +
                  mapping['E'] * 10 +
                  mapping['Y'])
        if SEND + MORE == MONEY:
            print("Solution Found!\n")
            print("Letter -> Digit Mapping")
```

```

    for letter in letters:
        print(f'{letter} = {mapping[letter]}')
    print("\nVerification")
    print(" SEND =", SEND)
    print(" MORE =", MORE)
    print("----- +")
    print("MONEY =", MONEY)
    return

print("No solution found.")
solve_cryptarithmic()

```

Output:

```

Output

Solution Found!

Letter -> Digit Mapping
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2

Verification
SEND = 9567
MORE = 1085
----- +
MONEY = 10652

=== Code Execution Successful ===

```

Result :

The Crypt Arithmetic Problem was successfully solved using the Backtracking Search algorithm. A unique digit was assigned to each letter while satisfying the equation:

$$\mathbf{9567 + 1085 = 10652}$$

Thus, the solution was obtained successfully.